

PROLOG PROGRAM – COUNT THE NUMBER OF VOWELS IN A WORD

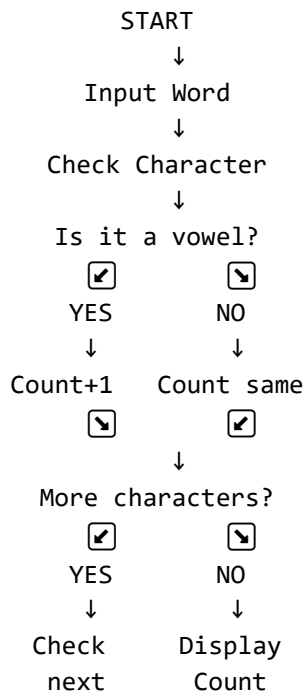
Aim

To write a Prolog program to count the number of vowels in a given word.

Algorithm

1. Start.
2. Accept a word as input.
3. Check each character of the word.
4. If the character is a vowel (a, e, i, o, u), increase the count by 1.
5. If it is not a vowel, keep the count unchanged.
6. Repeat until all characters are checked.
7. Display the total number of vowels.
8. Stop.

Flowchart

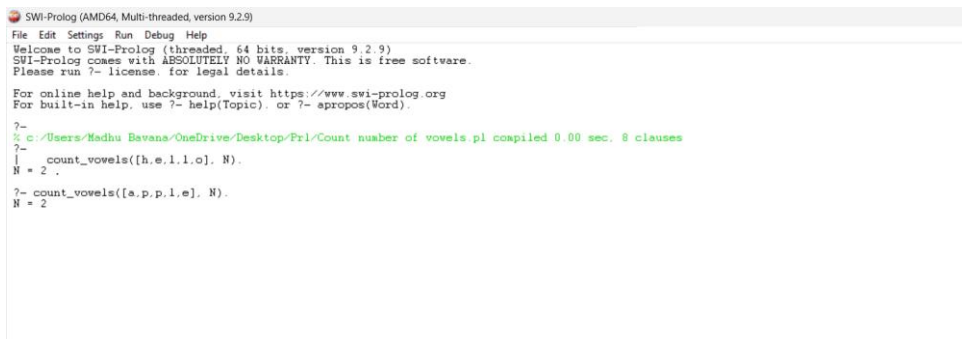


character ↓
STOP

Prolog Program

```
vowel(a).  
vowel(e).  
vowel(i).  
vowel(o).  
vowel(u).  
  
count_vowels([], 0).  
  
count_vowels([H|T], N) :-  
    vowel(H),  
    count_vowels(T, N1),  
    N is N1 + 1.  
  
count_vowels([H|T], N) :-  
    \+ vowel(H),  
    count_vowels(T, N).
```

OUTPUT



SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)

File Edit Settings Run Debug Help

Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit <https://www.swi-prolog.org>
For built-in help, use ?- help(Topic), or ?- apropos(Word).

```
?-  
% c:/Users/Madhu Bavana/OneDrive/Desktop/Frl/Count number of vowels.pl compiled 0.00 sec, 8 clauses  
?-  
| count_vowels([h,e,l,l,o], N).  
N = 2 .  
?- count_vowels([a,p,p,l,e], N).  
N = 2
```

Result

N = 2.

Therefore, the number of vowels in the word 'HELLO' is 2.